

Resilience of Cyber-Enabled Electrical Energy and Water Distribution Systems Considering Infrastructural Robustness Under Conditions of Limited Water and/or Energy Availability

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Abstract—New metrics and a new methodology for determining system infrastructural–operational resilience are presented for the optimal real-time operation of two critical interdependent infrastructure systems—the water distribution systems (WDS) and the electric power system (EPS) under critical conditions of limited water and/or limited electrical energy resulting from extreme drought or electric grid failure, respectively. Operational resilience (OR) and infrastructural resilience are defined. The integrated resilience computation method is presented and utilizes results from a combined optimization–simulation framework, which involves capturing the interactions and associated dynamics between the EPS and WDS. A realistic example of the WDS and EPS is used to demonstrate the application of the resilience concepts to assess the interdependent systems’ performance.

Index Terms—Electric power system (EPS), extended time-domain simulations, infrastructural resilience (IR), operational resilience (OR), real-time operations, water distribution systems (WDS), water energy nexus.

NOMENCLATURE

Terms

ceil	Ceiling function rounds up to the next highest integer.
$LOL_{n,t}$	Loss of load, bus n , time period t .

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$dAdj_{xf,n,t}$

$LOL_{n,t}$

mf_n

$P_{l,max}$

$P_{l,t}$

$P_{load,nt}$

sf_n

sl

$TL_{g,t}$

WDS

t

z

Z

$\theta_{z,t}$

$R_{P,low}^{n,tim}$

$R_{P,high}^{n,tim}$

$D_{sat,z,t}$

$D_{req,z,t}$

f

n

tim

$P_{n,t}$

$P_{s,low}$

P_{low}

$P_{s,upper}$

Load reduction amount, $xf \in \{sf, mf\}$, bus n , time t , from consumer survey results.

Loss of load, bus n , time period t .

Fraction of multifamily (mf) homes in residential component of load, bus n .

Rate A MVA limit, transmission element l .

Active power flow, transmission element l , time period t .

Real power demand, bus n , time period t .

Fraction of single-family (sf) homes in residential component of load at bus n .

Transmission line index function calculation parameter, summing length.

Cooling water storage tank level, generator g , time-period t .

Water distribution system.

Simulation time period.

Index for service area in WDS.

Total number of service areas in the system.

Penalty function for not satisfying demand for service area z at time t .

Binary indication function for low pressures.

Binary indication function for high pressures.

Satisfied demand multiplier for zone z at time t .

Requested demand multiplier for zone z at time t .

Index for performance function.

Index node in the WDS.

Index for real time of optimization–simulation implementation.

Pressure at node n and time t .

System safe lower pressure bound.

System lower pressure bound.

System safe upper pressure bound.

P_{upper}	System upper pressure bound.	$M_{EB,l}$	EPS betweenness metric for element l .
Pump	Index for pump number.	$M_{EFF,n}$	EPS efficiency metric for bus n .
$POW_{pump,t}$	Power consumed by pump at time t .	WDS IOR Functions	
$Pow(available)_{pump,t}$	Power available for consumption at time t .	$R_{R_P^{n,tim},t}$	Final resilience value for pressure, IOR-based calculation, and time t .
WDS Index Functions		$R_{R_{Pow}^{pump,tim},t}$	Final resilience value for pump power, IOR-based calculation, and time t .
$R_P^{n,tim}$	Pressure bounds index function, node n .	$R_{R_{Dem}^{z,tim},t}$	Final resilience value for demand satisfaction, IOR-based calculation, and time t .
$R_{Pow}^{pump,t}$	Power availability index function.	EPS IOR Functions	
$R_{Dem}^{z,t}$	Demand satisfaction index function, zone z .	$R_{R_V^n,t}$	Final resilience value for voltage, IOR-based calculation, and time t .
EPS Index Functions		$R_{R_{TL}^l,t}$	Final resilience value for transmission elements, IOR-based calculation, and time t .
R_{CW}^g	Cooling water index function, generator g .	$R_{R_{DS}^n,t}$	Final resilience value for overall demand satisfied, IOR-based calculation, and time t .
R_{DS}^n	Overall demand supplied index function, bus n .	$R_{R_{PL}^n,t}$	Final resilience value for pump demand supplied, IOR-based calculation, and time t .
R_{PL}^n	WDS pump demand supplied index function, bus n .	Total IOR Values	
R_{TL}^l	Transmission line thermal limit index function, branch l .	$R_{TOT}(t)$	Combined WDS and EPS IOR measure of performance.
R_V^n	Voltage index function, bus n .	$ReSEPS,IND$	EPS independent IOR terms: voltage, transmission line, and overall load supplied.
WDS Operational Resilience Measures of Performance (RMP)		$ReSEPS,INTR$	EPS interdependent IOR terms: thermoelectric cooling water supplied and WDS pump load supplied.
$Res_{R_P^{n,tim},t}$	RMP for pressure bounds function at time t .	$Res_{WDS,IND}$	WDS independent IOR term: pressure.
$Res_{R_{Pow}^{pump,tim},t}$	RMP for power availability function at time t .	$Res_{WDS,INTR}$	WDS interdependent IOR terms: pump power availability and demand satisfaction.
$Res_{R_{Dem}^{z,tim},t}$	RMP for demand satisfaction at time t .	IOR Quantification Metrics	
$Res_{WDS,TOT}(t)$	Overall WDS operational resilience at time t .	M_{ROB}	Metric quantifying system robustness to disturbance.
EPS Operational RMP		$M_{RAPI,DP}$	Metric quantifying system ability to limit the rate of impact by disturbance.
$Res_{R_V^n,t}$	RMP for voltage, time period t .	$M_{RAPI,RP}$	Metric quantifying system ability to rapidly recover from disturbance.
$Res_{R_{TL}^l,t}$	RMP for transmission lines, time period t .	M_{RA}	Metric quantifying system ability to reach predisturbance resilience value.
$Res_{R_{CW}^g,t}$	RMP for cooling water supply, time period t .	M_{TAPL}	Metric quantifying performance loss of system during disturbance.
$Res_{R_{DS}^n,t}$	RMP for overall demand supplied, time period t .	M_{TOT}	Metric quantifying overall system response to disturbance.
$Res_{R_{PL}^n,t}$	RMP for WDS pump demand supplied, time period t .	Weights	
$ReSEPS,TOT(t)$	Overall EPS operational resilience at time t .	w_{ov}, w_{otl}, w_{ocw}	Weights associated with voltage, transmission element, and cooling water.
WDS Infrastructural Robustness Metrics			
$M_{Conn,n}$	WDS connectivity metric for node n .		
$M_{bet,n}$	WDS betweenness metric for node n .		
$M_{demand,z}$	WDS demand priority metric for zone z .		
$M_{demAdj,z}$	WDS demand adjustment metric for zone z .		
EPS Infrastructural Robustness Metrics			
$M_{ED,n}$	EPS connectivity metric for bus n .		

w_{ods}, w_{opl}	Overall demand supplied and WDS pump demand supplied terms, respectively, for EPS OR,
$w_{op}, w_{opow}, w_{odem}$	Weights associated with pressure, pump power, and demand satisfaction terms, respectively, for WDS OR calculation,
w_1, w_2	Weights associated with the independent and interdependent terms, respectively, for IOR.
$w_v, w_{tl}, w_{cw}, w_{ds}, w_{pl}$	Weights associated with voltage, transmission element, cooling water, overall demand supplied, and WDS pump demand supplied terms, respectively, for EPS IOR.
w_p, w_{pow}, w_{dem}	Weights associated with pressure, pump power, and demand satisfaction terms, respectively, for WDS IOR.

I. INTRODUCTION

THE electric power system (EPS) and the water distribution system (WDS) are two operationally interdependent critical infrastructure systems that have been recognized and studied in broad generalities for some time. The main dependence of the EPS on the WDS is the water required for the cooling cycle of thermoelectric power generation. The main dependence of the WDS on the EPS is the electric power needed for pumping water from a source to the treatment plant and end-user through the WDS. The interdependencies become critical under conditions of limited water and/or electricity availability. The optimization–simulation framework [1] used for this article accounts for these interdependencies. If power plant cooling water demands are not met, then the EPS may fail to supply adequate electric power for WDS pumping stations leading to a cascading failure of both systems.

This article presents new metrics and a new methodology for computation of infrastructural–operational resilience (IOR) for real-time operations of WDS and EPS under critical conditions of limited water and/or limited electrical energy resulting from extreme drought or electric grid failure. IOR is calculated by weighting the operational resilience (OR) with the appropriate infrastructural resilience (IR) metrics. The methodology for the resilience calculations is depicted in Fig. 1 and considers the impacts of both reduced water availability and electric power supply. The computation presented involves calculations that capture the robustness of a systems’ structural configuration as well as the operating parameters.

A. Definition of OR and IR for WDS and EPS Systems

1) *Resilience of Critical Infrastructure Systems*: Increasing attention had been paid to resilience calculations for critical infrastructure systems. The US National Infrastructure Advisory Council broadly defines resilience [2] with four distinct characteristics in the context of critical infrastructure systems.

- 1) Robustness—encompasses the ability of a system to remain operable after contingencies occur.

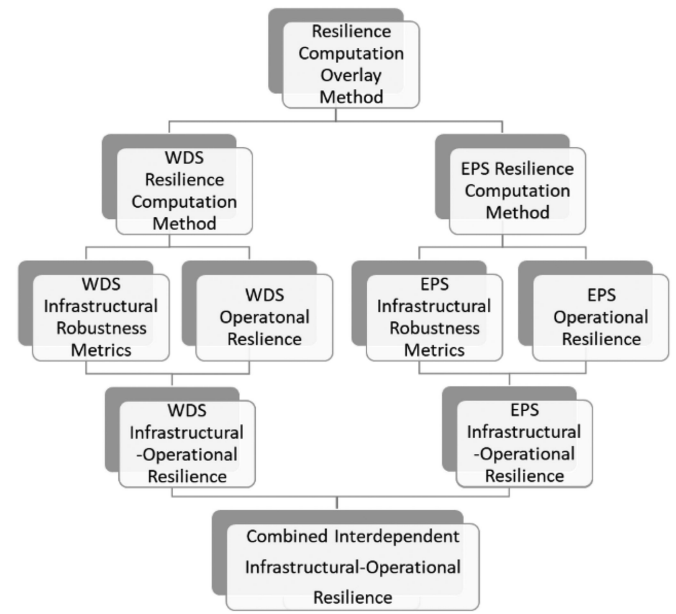


Fig. 1. Resilience computation methodology.

- 2) Resourcefulness—the ability to manage disasters.
- 3) Rapid recovery—the ability to manage disasters, limit the impact, and return to predisaster operating conditions.
- 4) Adaptability within the system and its operating procedures and policies to improve the robustness and recovery characteristics for future events.

A few notable definitions of resilience of an engineered system are given in [3]–[7]. The EPS and the WDS are inherently reliable in their operation and resilient to disturbances that impact the systems due to the extensive and thorough design and planning procedures of the two systems. However, metrics that capture interdependent quantities within each system and methods by which to calculate those metrics have not been previously explored.

2) *Resilience of the WDS*: OR of WDS is defined as how quickly the system recovers from not meeting acceptable pressure levels. Aydin *et al.* [8] defined OR as the ratio of the number of times a satisfactory performance follows an unsatisfactory performance and the number of times an unsatisfactory performance occurs within the system. A few definitions of resilience for WDS network structure and operations have been reported in the literature. In [6], [9]–[15], different definitions for WDS design and operations are presented. Shin *et al.* [16] investigated the definitions of quantitative resilience measures for WDSs in detail. Recently, a new methodology for the computation of OR for WDS has been presented [17]. Most of the studies in the literature focus on the resilience-based design of WDS except for a couple of studies in [8] and [18], which are based on OR of WDS. OR is defined as the capacity of an engineered system to absorb disturbances and reorganize while undergoing change so as to still retain essentially the same function, structure, identity, and feedbacks [5], [9]. This article uses the definition of OR given by Aydin *et al.* [8] for developing a real-time resilience assessment methodology for WDS operations under emergency conditions. An unsatisfactory performance would

result when one of the performance conditions, such as pressure requirements, is not met.

3) *Resilience of the EPS*: In [19], the potential benefits resulting from the enhancement of system performance in the face of extreme events are highlighted. Recent work in [20] has offered resilience metrics and a framework to quantify resilience in power systems. The authors presented measures of resilience that were characterized using the results of case studies of transmission line operation under high winds. Nan and Sansavini [21] provided similar metrics and a general framework to calculate overall system resilience of interdependent infrastructures from specified measures of performance.

4) *Behavior of RMP*: These measures of performance examine system behavior as functions of time before, during, and after system disturbances. The severity of a disturbance is measured by the degree to which the measure of performance drops, how fast it both drops and recovers, and to what value the measure settles at. These metrics are adequate to study the desired high impact, low probability extreme events that correspond directly with the terms outlined above for defining system resilience.

B. Previous Studies on Real-Time Operations of WDS and EPS

A few studies have reported on the real-time optimization of WDS and EPS controls considering water–energy nexus interactions. Fang and Chen [22] investigated the links between WDS and energy systems for a city. Vakilifard *et al.* [23] provided a detailed literature review for techniques and approaches for optimization of WDS considering water–energy nexus. An optimization model for real-time operations of water–energy nexus systems was presented in [24]. The model considered optimization of pump schedules during power outages but did not use a WDS simulation model. Oikonomou and Parvania [25] presented a methodology for optimal coordinated controls of WDS and EPS considering minimization of operational costs for both the systems. The study does not consider WDS operations for demand satisfaction under emergency conditions or simulation of real-time WDS operations. The methodology presented herein considers optimization–simulation of real-time operations of WDS under emergency conditions of limited power and/or water availability interfaced with an OR computation methodology for performance assessment.

II. OPTIMIZATION–SIMULATION APPROACH

The IOR methodology was applied using the results from an optimization–simulation model application developed for extended time-period WDS–EPS operations [1]. The power system control optimization was implemented by a combination of unit commitment (UC) for determining the ON/OFF status of all the generators and a modified dc optimal power flow (OPF) for determining generator output set-points. The power system network simulation for the operational time-frame simulations was accomplished by the use of time-series power flows. The water system optimization–simulation was accomplished using an extended period simulation coupled with a genetic algorithm (GA) solution methodology for a control optimization model.

The water system and the power system were interfaced by alternatively exchanging data between the power system and water

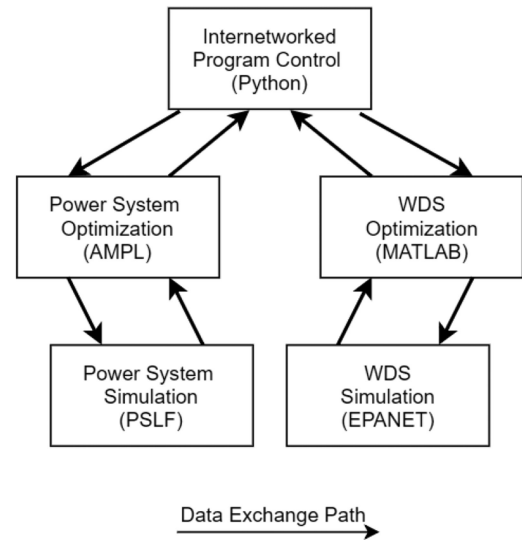


Fig. 2. Internet-worked simulation overview.

network simulations via a program control overlay developed in python. Fig. 2 depicts an overview of this concept. The relevant simulation data exchanged between the power system and water network simulations included EPS network configurations, pump electric power consumption as an identifiable component of power system loads, and thermal generation cooling water requirements, which were used as node demands within the water system. The temporal dependence of the WDS on the EPS is accomplished through the modification of the operating conditions in the EPS and vice versa. The spatial relationship between the two systems is accounted for by placing those loads or demands at the desired nodes or buses within the EPS and WDS.

All the models for the optimization and control of the EPS were solved via implementation with AMPL [26]. The complete EPS network solution that obtained the net active and reactive power injections as well as the voltage magnitudes and angles at each bus was obtained using GE's PSLF software [27]. The EPANET software [28] was used for the extended period simulation of the WDS being analyzed, whereas a MATLAB toolkit [29] was used to implement the genetic algorithm and to interface the WDS simulation with the optimization procedure. The objective of the optimization model (genetic algorithm) used for optimal real-time WDS operations is to minimize the difference between the requested demands ($D_{\text{req},z,t}$) and satisfied demands ($D_{\text{sat},z,t}$).

EPANET is a demand-driven simulation model that always meets the demands irrespective of pressures at nodes in the WDS. The solutions may have negative pressures at certain nodes for operations where enough water is not pumped to meet the water demands. For every time-step in the simulation, an iterative process is conducted to avoid such conditions. The iterative process consists of an optimization model solved using a GA providing a new set of decision variables (including satisfied demand multipliers and pump controls) to EPANET, and a simulation is performed based on these new decision variables. A negative pressure penalty function with a large penalty weight is used in the reduced optimization model to exclude

EPANET solutions with negative pressures. The final solution from the optimization–simulation can never include an operation schedule that results in negative pressures. A detailed description of the optimization–simulation methodology is given in [30].

The overall real-time operation of the WDS is described as follows.

- 1) Update all pump status from electrical energy input from EPS and update WDS input data (tank levels, status of valves, and flows).
- 2) Run the WDS optimization–simulation model to determine the pump operations and demand pattern given the scenario defined availability of water and electrical energy input from the EPS.
- 3) Provide the outputs of pump power consumption and cooling water flow to generators for the correct hour to the EPS.

The overall real-time operation of the EPS is described as follows.

- 1) Update topology (bus and branch status) from the scenario contingency definition.
- 2) Update WDS-related quantities (pump loads and generator node water flows).
- 3) Optimize the EPS model to determine the UC and unit dispatch using the results from 1 and 2.
- 4) Calculate updated cooling water requirements and make policy adjustments.
- 5) Provide pump nodes electric supply availability and new generator node cooling water demands to the WDS.

III. OR INDEX FUNCTIONS FOR WDS AND EPS

OR is a weighted metric that accounts for the pressure, pump power, and demand satisfaction for WDS; voltage, transmission line thermal limits, thermoelectric generation cooling water, and load supply for the EPS. Index functions are defined as indices that allow for the current system conditions to be gauged in terms of the selected operational parameters. The index functions for both WDS and EPS are defined in this section.

A. WDS: Pressure Function

Pressure bounds are an important performance assessment criterion for WDS operations. During emergency conditions, water demands may be satisfied at lower pressures than generally acceptable. Lower pressure bound violations represent a partial failure of the WDS since prolonged lower pressure bound violations could be a public health safety hazard. Similarly, water demands satisfied at pressures higher than the upper pressure bounds may lead to failure of WDS components or household faucets. Therefore, from an OR perspective, satisfaction of pressure bounds is an important consideration for the demands that can be satisfied.

A low-pressure function $R_{P,low}^{n,tim}$ has a value of 1 when node pressures are above the safe lower limit of $P_{s,low}$ and a value of 0 for pressures lower than the extreme limit P_{low} . The pressure function is allocated a value (between 0 and 1) based on its deviation from the $P_{s,low}$, between P_{low} and $P_{s,low}$. The decision conditions for low-pressure indication function for nodes n and

simulation time tim are as follows:

$$\begin{aligned} &\text{If } P_{n,tim} \leq P_{low}, \text{ then } R_{P,low}^{n,tim} = 0 \\ &\text{Else, if } P_{n,tim} > P_{low} \text{ and } P_{n,tim} \leq P_{s,low}, \\ &\text{then } R_{P,low}^{n,tim} = \min \left(0, \left(\frac{P_{n,tim} - P_{low}}{P_{s,low} - P_{low}} \right) \right). \end{aligned} \quad (1)$$

For pressures higher than the safe limit of $P_{s,upper}$, a high-pressure indication function $R_{P,high}^{n,tim}$ has a value of 1 when node pressures are below $P_{s,upper}$, whereas the function value is 0 for pressures higher than the extreme limit of P_{upper} . The pressure function is allocated a value between (0 and 1) based on its deviation from the safe limit of $P_{s,upper}$ if the pressure lies between $P_{s,upper}$ and P_{upper} . The decision conditions for high-pressure indication function ($R_{P,high}^{n,tim}$) for nodes n and simulation time tim are as follows:

$$\begin{aligned} &\text{If } P_{n,tim} \geq P_{upper} \text{ then } R_{P,high}^{n,tim} = 0 \\ &\text{Else, if } P_{n,tim} \geq P_{s,upper} \text{ and } P_{n,tim} < P_{upper}, \\ &\text{then } R_{P,high}^{n,tim} = \max \left(0, \left(\frac{P_{upper} - P_{n,tim}}{P_{upper} - P_{s,upper}} \right) \right). \end{aligned} \quad (2)$$

A combined function $R_P^{n,tim}$ for pressures is defined as follows:

$$\begin{aligned} &\text{If } R_{P,low}^{n,tim} = 0 \text{ or } R_{P,high}^{n,tim} = 0, \text{ then } R_P^{n,tim} = 0 \\ &\text{Else, } R_P^{n,tim} = \frac{R_{P,low}^{n,tim} + R_{P,high}^{n,tim}}{2}. \end{aligned} \quad (3)$$

B. WDS: Power Availability Function

An important aspect of scheduling pump operations is to avoid electric power availability upper bound violations. For improved OR, compliance with the electric power availability is one of the major contributing factors. Each pumping station in the WDS is equipped with a back-up power generator system (primarily diesel generation). The optimization–simulation model used for WDS pump controls penalizes use of these generators due to reliability concerns during emergency conditions. The aim of the control optimization is to operate the WDS pumps within the available power from the PDS to meet the water demands. The function for pump power, $R_{Pow}^{pump,t}$, for all pumps and times t is considered to be binary (0, 1). The function has a value of 1 for the pumps complying with the power availability. The function assumes a value of 0 when the power availability is not satisfied. The decision rule for the power compliance function is given by

$$\begin{aligned} &\text{If } Pow_{pump,t} > Pow(\text{available})_{pump,t}, \text{ then } R_{Pow}^{pump,t} = 0 \\ &\text{If } Pow_{pump,t} \leq Pow(\text{available})_{pump,t}, \text{ then } R_{Pow}^{pump,t} = 1. \end{aligned} \quad (4)$$

C. WDS: Demand Satisfaction Function

The function for demand satisfaction is given by $\frac{D_{sat,z,t}}{D_{req,z,t}}$, which is an output from the optimization–simulation model. The values of the function show the extent to which the requested

TABLE I
PARAMETERS FOR BUS VOLTAGE RESILIENCE INDEX FUNCTION

Symbol	Quantity	Value (pu)
V_{min}	Lower-Cut Out for Index Function	0.900
$V_{thr,min}$	Minimum Bus Voltage Value for Maximum Index Function	0.975
$V_{thr,max}$	Maximum Bus Voltage Value for Maximum Index Function	1.025
V_{max}	Upper-Cut Out for Index Function	1.100

demands are satisfied. The function for demand satisfaction is given as

$$R_{Dem}^{z,t} = \frac{D_{sat,z,t}}{D_{req,z,t}}. \quad (5)$$

D. EPS Introduction

The following quantities are used for calculating the OR index function values:

- 1) bus voltage magnitude;
- 2) line thermal limits;
- 3) *thermoelectric cooling water demand satisfaction*;
- 4) load (overall) supply satisfaction;
- 5) *load (WDS pump) supply satisfaction*.

The terms in italics above denote quantities that capture resilience in terms of explicit interdependencies between the two systems. Each of these five quantities is evaluated for every iteration (time period) within the simulation.

E. EPS: Voltage Index Function

Power system bus voltage magnitudes are maintained within a small range of values during normal system operation. Because electrical equipment is designed for operation at certain rated voltages, therefore operation under conditions that differ greatly from the rated specifications can have adverse effects. These effects can include insulation breakdown for equipment that is operated at voltage levels that are too high. The index function relating the operational performance of the power system to bus voltages is therefore defined to have a value of 1 for voltages within a small, nominal operating range. Outside of this range, the resilience is assumed to decrease as the deviation away from the nominal operating conditions is increased and reach 0 at some point. Following the description mentioned above and using the values for thresholds, minimum, and maximum voltages given in Table I, the resilience index function for bus voltage magnitude is defined mathematically as follows (in terms of a piecewise linear function describing the thresholds in Table I):

$$R_V^n = \begin{cases} 0 & V_n < V_{min} \\ 13.3 \cdot V_n - 12 & V_{min} \leq V_n \leq V_{thr,min} \\ 1 & V_{thr,min} \leq V_n \leq V_{thr,max} \\ -13.3 \cdot V_n + 14.67 & V_{thr,max} \leq V_n \leq V_{max} \\ 0 & V_{max} < V_n \end{cases} \quad (6)$$

F. EPS: Transmission Line Thermal Limit Index Function

Similar to the voltage magnitudes in the power system, the transmission lines are designed to be operated under specific rated current magnitude levels. The reason for this is that the equipment, consisting of conductors, terminating equipment, such as transformers, and protection equipment, all have thermal ratings and operation of these equipment above these temperatures can cause permanent damage to some of the most expensive components in the power system. In addition, transmission lines will sag under increasing current (and therefore temperatures), which can cause regulatory issues regarding minimum clearance criteria as well as pose a potential hazard for objects beneath the transmission lines.

Because power is proportional to current, lines are normally given several different maximum apparent power operating limits. These include the familiar normal, continuous Rate A MVA equipment rating as well as the emergency, Rate C MVA limit. Because there is flexibility on the magnitude of power flow on a line, a moving average of the line flow is calculated and used for the transmission line thermal limit index function, defined as

$$R_{TL}^l = \min \left(1, \frac{P_{l,max} \cdot sl}{\sum_{r=t-sl}^{t-1} P_{l,r}} \right). \quad (7)$$

G. EPS: Thermoelectric Generation Cooling Water Index Function

Thermoelectric generation requires water for the cooling cycle in the process of converting the heated working fluid from steam back to water via heat transfer. The closed-loop cooling cycle introduces losses through drift, blowdown, and evaporation. So, water is needed to “make-up” for the amount lost and represents a key dependence of the power system on the water system. Generation owners meet this need through external sources, such as a municipal water utility as well as by water that is stored onsite. Since the amount of onsite storage for cooling water is normally substantial [15], it is reasonable to assume that the levels of the water in the onsite tanks will not change vary rapidly. The index function relating system performance to the amount of cooling water that is being supplied is related to a conservative estimate of the storage tank level with a piecewise linear form as follows:

$$R_{CW}^g = \frac{\text{ceil} \left(\frac{TL_{g,t}}{TL_{g,initial}} \cdot 10 \right)}{10}. \quad (8)$$

This formulation of the index function gives a slight lag in the function’s value with respect to changes in the water storage

TABLE II
SUBCATEGORIES SURVEYED FOR USAGE IN EPS AND WDS DOMAINS

EPS Subcategory	WDS Subcategory
Cooling	Toilet
Dryer	Shower
Washing Machine	Washing Machine
Oven	Bath
Dishwasher	Dishwasher
Water Heater	Outdoor
Refrigeration	Leaks
Lighting	Faucet
Outdoor	Other
Cooking	-
Other	-

tank level. This characterizes the desired behavior of tracking the tank level but also ignoring small fluctuations in tank level.

H. EPS: Load Supplied (Overall) Index Function

The goal of power system operation is to satisfactorily supply the requested electrical power demands of consumers while satisfying constraints related to the quality of the power delivered. A resilience measure along these lines is formulated in order to quantify how well the system demands were met under extreme system operating conditions. Based on consumer surveys [31], consumers are willing to adjust their demands depending on the perceived weather conditions. These consumers are also observed to have different levels of willingness to change their consumption based on the knowledge of how the water and power systems interact as well as whether the curtailment of the usage of water and energy is mandatory or not. For representation of this fact within this resilience calculation, the bus loads are decomposed into representative percentages for industrial, commercial, and residential components. Within the residential load component, it is further subdivided into single-family (sf) and multifamily (mf) elements based on recent census data for Arizona [32].

An example of how incorporation of this consumer data will affect the load curve over time is demonstrated below. The consumer survey data regarding the change in usage of water and electricity was collected for subcategories of each domain, as shown in Table II. These subcategories represent the major end-uses of electricity and water for the residential sector in the United States. A scenario corresponding to a chronic water shortage was then selected from the consumer survey results, and the proportion of electricity usage that consumers would be willing to curtail is used to calculate the amount that would still be used. This was then weighted against a given subcategory's proportion of the total energy use in the US and aggregated in order to give the demand adjustment factors for the peak hours surveyed, as shown in Table III. The interpretation of these demand adjustment factors is that consumers would be willing

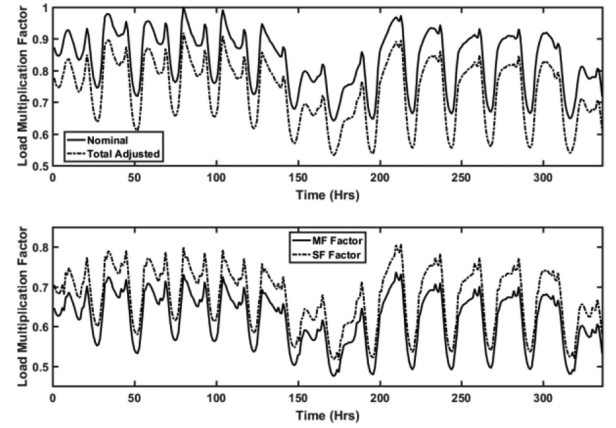


Fig. 3. Demand adjustment factor LMF curves.

to reduce their overall consumption by a factor equal to these values in the corresponding hour.

These values were then used to alter the nominal value of load multiplier factors (LMF) at any given time period for the load buses. The new LMF for the single-family or multifamily components of the load is given by the product of the nominal LMF curve and the demand coefficient (*sf* or *mf*) at that hour. For the hours that are not peak hours and hence no demand coefficient is given, an estimate that the demand adjustment is equal to the average of the peak hours was used. The plots at the top of Fig. 3 show the nominal LMF and the adjusted LMF, assuming percentages of single-family and multifamily houses as stated above and assuming that residential loads are one-third of the total load. The plot at the bottom of Fig. 3 shows the LMF factor corresponding to multifamily and single-family homes.

With the previous values defined and examined, the resilience index function for overall load satisfaction can now be defined and is expressed as follows:

$$\begin{aligned}
 R_{DS}^n &= \text{ceil}((P_{\text{load},nt} - \text{LOL}_{n,t}) \\
 &\quad - (P_{\text{load},nt} - \text{sf}_n \cdot d \text{Adj}_{\text{sf},t} - \text{mf}_n \cdot d \text{Adj}_{\text{mf},t})) \\
 &= \text{ceil}(\text{sf}_n \cdot d \text{Adj}_{\text{sf},t} + \text{mf}_n \cdot d \text{Adj}_{\text{mf},t} - \text{LOL}_{n,t}). \quad (9)
 \end{aligned}$$

The first term is the present bus load less any load that is lost and the second term is a best case load assuming that the present demand is adjusted by a decrease in value equal to the reduction in consumption by consumers. From this, the function is assigned a value of 1 if the actual load (less any lost) is greater than the best case consumer adjusted load and zero if the opposite is true.

I. EPS: Load Supplied (WDS Pumps) Index Function

In an effort to further capture the operational interdependencies between the water and electric systems, an additional index function is defined, which is related to the satisfaction of the pump load directly serving the WDS. In contrast to the previous load metric, this metric does not consider any consumer survey data but instead directly generates a one or a zero based on

TABLE III
PEAK HOUR DEMAND COEFFICIENTS FOR SINGLE-FAMILY AND MULTIFAMILY HOMES

Time	Demand Coefficient - <i>sf</i>	Demand Coefficient - <i>mf</i>
6am - 6:59am	0.75444906	0.7093457
7am - 7:59am	0.79997816	0.7314116
8am - 9am	0.78636656	0.7292784
5pm - 5:59pm	0.8309073	0.7605177
6pm - 6:59pm	0.82466484	0.7514394
7pm - 7:59pm	0.8223109	0.7490517
8pm - 9pm	0.82880098	0.7518957

whether the pump demand is satisfied or not. The resilience function is mathematically given as follows:

$$R_{PL}^n = \begin{cases} 1, & PL_{\text{demand}} = PL_{\text{supply}} \\ 0, & PL_{\text{demand}} > PL_{\text{supply}} \end{cases} \quad (10)$$

where PL_{demand} is the active power demand that would have been requested, and PL_{supply} is the value determined by the scenario definition (e.g., $PL_{\text{demand}} = PL_{\text{supply}}$ with no contingency).

IV. COMPUTATION OF OPERATIONAL RESILIENCE

The definition for OR for an engineered system for each index function f in the WDS and EPS at time t is calculated as

$$Res_{f,t} = \frac{\# \text{ of times satisfactory follows unsatisfactory}}{\# \text{ of times unsatisfactory occurs}}. \quad (11)$$

This article uses this definition of OR for deriving the equations for OR under emergency conditions. Unsatisfactory operation is defined as the resilience index function value being less than that function's value in the previous simulation hour (and satisfactory operation denoted by an increase in the index function's value). The measure of performance is defined for real values between 0 and 1.

A. WDS: Operational System Resilience

The overall resilience for time t is a normalized (real value between 0 and 1) weighted sum of the pressure, power, and demand resiliencies

$$Res_{WDS,TOT}(t) = \frac{(w_{op} Res_{R_P^{n,tim},t} + w_{opow} Res_{pow,t} + w_{odem} Res_{demands,t})}{(w_{op} + w_{opow} + w_{odem})} \quad (12)$$

where w_{op} , w_{opow} , and w_{odem} are weights associated with pressure, power, and demand satisfaction resilience, respectively. The values of these weights are determined using sensitivity analyses for a WDS. The weights (w_{op} , w_{opow} , and w_{odem}) are determined through a sensitivity analysis for each application of the IOR methodology. The sensitivity analyses are performed for all terms involved in WDS IOR computation. The value of weights to be used for a particular term depend upon the relative importance of the penalty term and the magnitude of the numeric value of the term. Parameters considered for the

sensitivity analyses are the value of the objective function, number of pressure bound violations, number of power bound violations, and the number of switch constraint violations. The optimization model is tested for a range of values for each of the penalty weights and a combination of penalty weights yielding the minimum objective function value, and least amount of bound violations for pressures, power, and water quality is chosen for the application.

B. EPS: Operational Resilience

The overall system resilience at time t is then calculated as a normalized weighted sum of the voltage, thermal limit, cooling water, demand supplied (overall), and demand supplied (WDS pumps) resilience values

$$Res_{EPS,TOT}(t) = \frac{(w_{ov} Res_{R_V^{n,t}} + w_{otl} Res_{R_{TL}^{n,t}} + w_{ocw} Res_{R_{CW}^{n,t}} + w_{ods} Res_{R_{DS}^{n,t}} + w_{opl} Res_{R_{PL}^{n,t}})}{(w_{ov} + w_{otl} + w_{ocw} + w_{ods} + w_{opl})} \quad (13)$$

w_v , w_{tl} , w_{cw} , w_{ds} , and w_{pl} are weights associated with voltage, transmission elements, cooling water demand satisfactions, overall demand satisfied, and WDS pump demand satisfaction, respectively.

C. WDS: Pressure Resilience

Resilience for the upper and lower pressure bound function ($R_P^{n,tim}$) defined in (3) is formulated using (11). The resilience for pressure bound satisfaction function at time t is given as follows:

$$Res_{R_P^{n,tim},t} = \frac{\left(\sum_n \sum_{tim} \max(0, R_P^{n,tim} - R_P^{n,tim-1}) \right)}{\left(\# \text{ of times } R_P^{n,tim} < 1 \right)_t} \quad (14)$$

$R_P^{n,tim}$ is the function of pressure bound satisfaction defined in (2) and (3) for simulation time tim and node n .

D. WDS: Power Consumption Resilience

Resilience for power availability function ($R_{Pow}^{pump,t}$) defined in (4) is formulated using (11). The resilience for power

availability function at time t is given as follows:

$$\text{Res}_{R_{\text{Pow}}^{\text{pump},\text{tim}},t} = \left(\sum_{\text{pump}} \sum_{\text{tim}}^T \max(0, R_{\text{Pow}}^{\text{pump},\text{tim}} - R_{\text{Pow}}^{\text{pump},\text{tim}-1}) \right) / \left(\# \text{ of times } R_{\text{Pow}}^{\text{pump},\text{tim}} = 0 \right)_t \quad (15)$$

$R_{\text{Pow}}^{\text{pump},\text{tim}}$ is the function of power availability defined in (4) for simulation time tim and pump.

E. WDS: Demand Satisfaction Resilience

Resilience for demand satisfaction function ($R_{\text{Dem}}^{z,t}$) defined in (5) is formulated using (11). The resilience for demand satisfaction function at time t is given as follows:

$$\text{Res}_{R_{\text{Dem}}^{z,\text{tim}},t} = \left(\sum_z \sum_{\text{tim}}^T \max(0, R_{\text{Dem}}^{z,\text{tim}} - R_{\text{Dem}}^{z,\text{tim}-1}) \right) / \left(\# \text{ of times } R_{\text{Dem}}^{z,\text{tim}} < 1 \right)_t \quad (16)$$

$R_{\text{Dem}}^{z,\text{tim}}$ is the function of demand satisfaction at simulation time tim and zone z .

F. EPS: Voltage Resilience

The resilience measure of performance for bus voltages is formulated using (11) and the bus voltage index function (R_V^n) given in (6) for each time t of the simulation and is calculated as follows:

$$\text{Res}_{R_V^n,t} = \left(\sum_{n=1}^{\text{NB}} \max(0, R_V^n(t) - R_V^n(t-1)) \right) / \left(\sum_{n=1}^{\text{NB}} \text{ceil}(R_V^n(t-1) - R_V^n(t)) \right) \quad (17)$$

$R_V^n(t)$ is the voltage magnitude resilience index function at bus n in time period t .

G. EPS: Transmission Line Thermal Limit Resilience

The measure of performance for transmission line thermal limits is formulated using (11) and the thermal limit index function (R_{TL}^l) given in (7) for each time t of the simulation and is calculated as follows:

$$\text{Res}_{R_{\text{TL}}^l,t} = \left(\sum_{l=1}^{\text{NL}} \max(0, R_{\text{TL}}^l(t) - R_{\text{TL}}^l(t-1)) \right) / \left(\sum_{l=1}^{\text{NL}} \text{ceil}(R_{\text{TL}}^l(t-1) - R_{\text{TL}}^l(t)) \right) \quad (18)$$

$R_{\text{TL}}^l(t)$ is the thermal limit resilience index function of element l in time-period t .

H. EPS: Thermoelectric Generation Cooling Water Resilience

The measure of performance for the thermoelectric generation's cooling water supply is then formulated using (11) and the cooling water index function (R_{CW}^g) given in (8) value for each time t in the simulation as follows:

$$\text{Res}_{R_{\text{CW}}^g,t} = \left(\sum_{g=1}^{\text{NG}} \max(0, R_{\text{CW}}^g(t) - R_{\text{CW}}^g(t-1)) \right) / \left(\sum_{g=1}^{\text{NG}} \text{ceil}(1 - R_{\text{CW}}^g(t)) \right) \quad (19)$$

$R_{\text{CW}}^g(t)$ is the thermoelectric generation cooling water index function for the generator g in time period t . In summary, this formulation quantifies the resilience with respect to cooling water as a piecewise linear function based on the ratio between the present onsite cooling water to the initial tank level (based on a two week supply).

I. EPS: Load Supplied (Overall) Resilience

The resilience measure of performance related to the supplied demand at time t is then formulated using (11) and the overall load supplied index function (R_{DS}^n) given in (9) and can be calculated as follows:

$$\text{Res}_{R_{\text{DS}}^n,t} = \left(\sum_{n=1}^{\text{NBL}} \max(0, R_{\text{DS}}^n(t) - R_{\text{DS}}^n(t-1)) \right) / \left(\sum_{n=1}^{\text{NBL}} \text{ceil}(R_{\text{DS}}^n(t-1) - R_{\text{DS}}^n(t)) \right) \quad (20)$$

J. EPS: Load Supplied (WDS Pumps) Resilience

The resilience measure of performance related to the supplied demand of pump loads in the WDS at time t is then formulated using (11) and the WDS pump load supplied index function (R_{PL}^n) given in (10) and can be calculated as follows:

$$\text{Res}_{R_{\text{PL}}^n,t} = \left(\sum_{n=1}^{\text{NPL}} \max(0, R_{\text{PL}}^n(t) - R_{\text{PL}}^n(t-1)) \right) / \left(\sum_{n=1}^{\text{NPL}} \text{ceil}(1 - R_{\text{PL}}^n(t)) \right) \quad (21)$$

and $R_{\text{PL}}^n(t)$ is the overall load supplied index function of pump load n in time period t .

V. INFRASTRUCTURAL ROBUSTNESS METRICS

Robustness is the inherent capability of a system to resist failures. This capability is an intrinsic property of a system based on its structure and configuration. This article considers metrics for computation of the system level robustness based on system structural configuration. The robustness metrics for the WDS and EPS are defined in the following sections. The EPS metrics are based upon those defined in [33] and [34].

A. WDS: Connectivity Metric

Connectivity of a node in a WDS is defined as the number of links (l_n) that are connected to a node n and characterizes the system interconnection and the degree of redundancy of the system components. The connectivity metric is given as

$$M_{\text{Conn},n} = \text{card}(l_n) \quad (22)$$

where $M_{\text{Conn},n}$ is the number of links connected to a node n , and l_n is the set of links (including pipes, pumps, and valves) connected to node n .

B. WDS: Betweenness Metric

The betweenness metric gives the robustness associated with the functionality of the links connected to a certain node. This metric is based on the type (pipe, pump, or valve) and size (diameter for pipe/valves or flow for pumps) of the connecting links. The betweenness metric scores the nodes based on the functional importance of the connecting links for WDS operations. A functionality factor (α_l) for link l is defined as a function of the link's type and size. For pipes, α_l is based on the size of the pipe and is calculated as the ratio of the diameter of pipe l divided by the maximum diameter of all L pipes in the WDS. For pumps, α_l is based on the maximum possible flow from the pump and given as the ratio of the flow of a given pump divided by the maximum flow in any of the WDS pumps. The betweenness metric for a node n in a WDS is derived as

$$M_{\text{bet},n} = \sum_{l(n)} \alpha_l \quad (23)$$

where $M_{\text{bet},n}$ is the betweenness metric for node n .

C. WDS: Demand Priority Metric

Demand priority is an important robustness parameter during extreme water shortage and limited electric power availability conditions. A demand pattern is defined for each service area in a WDS. The demand priority metric is simply a priority weight given to each service area z to ensure robustness with respect to demand satisfaction at higher priority service areas. The demand priority metric ($M_{\text{demand},z}$) values are assigned a noninteger value between 0 and 1 for each service area in the system depending on the type of demands (residential, commercial, industrial, or cooling water for power plants) in the service area.

D. WDS: Demand Adjustment Metric

Satisfaction of water demands in a WDS is an important requirement during normal and emergency operating conditions. For extreme drought and electrical outage scenarios, a part of the water demand at certain service areas may not be satisfied. In such conditions, the willingness of the consumers to reduce the water consumption helps in operating the WDS to supply the required water to priority locations, such as hospitals, public utilities, or fire stations. The demand adjustment metric ($M_{\text{dem Adj},z}$) gives the extent to which the population in a service area is willing to adjust their demands in an event of short- and long-term water and electric power shortage. This metric

is derived from extensive surveys conducted in Arizona for single-family units and multifamily units [31]. For every service area z in the system, two metrics are defined for single-family units ($dAdj_{\text{sf},t}$) and multifamily units ($dAdj_{\text{mf},t}$) based on the proportion of single-family units and multifamily units present.

E. EPS: Connectivity Metric

The connectivity metric for a bus is determined by the impedance of the lines connected to it. This metric will be used as a weight in order to properly account for the importance of an individual bus within the overall bus voltage resilience calculation. The metric is defined as follows:

$$M_{\text{ED},n} = \left(1 - \sum_m p_{nm} \cdot \log(p_{nm}) \right) \sum_m w_{nm} \quad (24)$$

where the weight of the edge, w_{nm} , is the weight of a transmission line (edge) and is assigned a value equal to the primitive impedance of the transmission element, and p_{nm} is defined as the weight w_{nm} divided by the sum of all weights.

F. EPS: Betweenness Metric

Similar to the connectivity measure above but for a transmission line (edge) rather than a bus (node), the betweenness metric captures the importance of a given transmission line in the resilience calculation for thermal limits. The calculation for the extended betweenness for a transmission element l is given as follows:

$$M_{\text{EB},l} = \left| \sum_g \sum_d C_{g,d} \text{PTDF}_{l,g,d} \right| \quad (25)$$

where $C_{g,d}$ is the minimum ratio for all lines of the quantity $P_{\text{max},l}$, the maximum active power across transmission element l , and the power transfer distribution factor from generator g to load d across line l .

G. EPS: Efficiency Metric

The efficiency metric uses the electrical distance between generators and a load to determine how important a given load is in the calculation of its resilience value. Defining the equivalent impedance between generator g and load n as follows:

$$Z_g^n = z_{gg} - 2z_{gn} + z_{nn} \quad (26)$$

where the z_{ij} values are the ij th entry in the bus impedance matrix, and the efficiency metric for the load at bus n can be calculated as follows:

$$M_{\text{EFF},n} = \frac{1}{N_g} \sum_g \frac{1}{Z_g^n}. \quad (27)$$

VI. IOR COMPUTATIONS

A. WDS: Pressure IOR

The pressure IOR considering infrastructural robustness metrics for connectivity and betweenness at time t is given as

follows:

$$R_{R_P^{n,\text{tim}},t} = \left(\sum_n^N \sum_{\text{tim}}^T \left[\left(\frac{M_{\text{Conn},n}}{\sum_n^N M_{\text{Conn},n}} \right) \left(\frac{M_{\text{bet},n}}{\sum_n^N M_{\text{bet},n}} \right) \right] \right) \times \max \left(0, R_P^{n,\text{tim}} - R_P^{n,\text{tim}-1} \right) \Bigg/ \left(\# \text{ of times } R_P^{n,\text{tim}} < 1 \right)_t. \quad (28)$$

B. WDS: Power Consumption IOR

The functionality factor for pumps (α_{pump}) is considered for computing the IOR for power consumption at time t as follows:

$$R_{R_{\text{Pow}}^{\text{pump},\text{tim}},t} = \left(\sum_{\text{pump}}^{\text{PUMP}} \sum_{\text{tim}}^T \left[\left(\frac{\alpha_{\text{pump}}}{\sum_{\text{pump}}^{\text{PUMP}} \alpha_{\text{pump}}} \right) \right] \right) \times \max \left(0, R_{\text{Pow}}^{\text{pump},\text{tim}} - R_{\text{Pow}}^{\text{pump},\text{tim}-1} \right) \Bigg/ \left(\# \text{ of times } R_{\text{Pow}}^{\text{pump},\text{tim}} = 0 \right)_t. \quad (29)$$

C. WDS: Demand Satisfaction IOR

Demand priority and demand adjustment metrics are considered for computation of the demand satisfaction IOR. A mathematical function $g\{\text{demand}_{z,t}\}$ is used to determine whether the satisfied demand (D_{sat}) is within the demand adjustment metric for a certain zone. The mathematical function is given as follows:

$$g\{\text{demand}_{z,t}\} = \max \left(0, \frac{D_{\text{sat},z,t}}{D_{\text{req},z,t}} - \text{mfn}_z d \text{Adj}_{\text{mf},t} - \text{sfn}_z d \text{Adj}_{\text{sf},t} \right) \quad (30)$$

where mfn_z and sfn_z are ratios of multifamily units and single-family units in service area z , respectively.

The computation for demand satisfaction resilience is as follows:

$$\begin{aligned} \text{if } g\{\text{demand}_{z,t}\} = 0, \quad \text{then } ds_{z,t} &= 0 \\ \text{if } g\{\text{demand}_{z,t}\} > 0, \quad \text{then } ds_{z,t} &= \frac{D_{\text{sat},z,t}}{D_{\text{req},z,t}}. \end{aligned} \quad (31)$$

The IOR for WDS demand satisfaction is given as

$$R_{R_{\text{Dem}}^{z,\text{tim}},t} = \sum_z^Z \left(\frac{M_{\text{demand},z}}{\sum_z^Z M_{\text{demand},z}} \right) ds_{z,t}. \quad (32)$$

D. WDS: System IOR

The total system IOR is computed as a weighted sum average of the infrastructural–operational resiliencies for pressure, power consumption, and demand satisfaction as follows:

$$R_{\text{WDS,TOT}}(t)$$

$$= \frac{w_p R_{R_P^{n,\text{tim}},t} + w_{\text{pow}} R_{R_{\text{Pow}}^{\text{pump},\text{tim}},t} + w_{\text{dem}} R_{R_{\text{Dem}}^{z,\text{tim}},t}}{w_p + w_{\text{pow}} + w_{\text{dem}}}. \quad (33)$$

E. EPS: Voltage IOR

Resilience for bus voltages in the IOR context is calculated by weighting the operation bus voltage resilience value with the connectivity metric expressed as follows:

$$R_{R_V^n,t} = \left(\sum_{n=1}^{\text{NB}} \left(\left(1 - \sum_m p_{nm} \cdot \log(p_{nm}) \right) \cdot \sum_m w_{nm} \right) \cdot \max(0, R_V^n(t) - R_V^n(t-1)) \right) \Bigg/ \left(\sum_{n=1}^{\text{NB}} \text{ceil}(R_V^n(t-1) - R_V^n(t)) \right). \quad (34)$$

F. EPS: Transmission Line Thermal Limit IOR

The IOR value for transmission line thermal limit resilience is calculated by weighting the operational thermal limit resilience with the extended betweenness metric

$$R_{R_{\text{TL}}^l,t} = \left(\sum_{l=1}^{\text{NL}} \left(\left| \sum_g \sum_d C_{g,d} \text{PTD } F_{l,g,d} \right| \right) \max(0, R_{\text{TL}}^l(t) - R_{\text{TL}}^l(t-1)) \right) \Bigg/ \left(\sum_{l=1}^{\text{NL}} \text{ceil}(R_{\text{TL}}^l(t-1) - R_{\text{TL}}^l(t)) \right). \quad (35)$$

G. EPS: Load Supplied IOR

Both the load supplied operational resilience values (overall and WDS pumps) are weighted at each node according to the proposed electrical efficiency metric

$$R_{R_{\text{DS}}^n,t} = \left(\sum_{n=1}^{\text{NBL}} \left(\frac{1}{N_g} \sum_g \frac{1}{Z_g^n} \right) \max(0, R_{\text{DS}}^n(t) - R_{\text{DS}}^n(t-1)) \right) \Bigg/ \left(\sum_{n=1}^{\text{NBL}} \text{ceil}(R_{\text{DS}}^n(t-1) - R_{\text{DS}}^n(t)) \right) \quad (36)$$

for the load supplied (overall) IOR and

$$R_{R_{\text{PL}}^n,t} = \left(\sum_{n=1}^{\text{NPL}} \left(\frac{1}{N_g} \sum_g \frac{1}{Z_g^n} \right) \max(0, R_{\text{PL}}^n(t) - R_{\text{PL}}^n(t-1)) \right) \Bigg/ \left(\sum_{n=1}^{\text{NPL}} \text{ceil}(1 - R_{\text{PL}}^n(t)) \right) \quad (37)$$

for the load (WDS pumps) IOR.

H. EPS: System IOR

The overall EPS IOR, $R_{\text{EPS,TOT}}(t)$, is then calculated at each time period t as a normalized weighted sum of the voltage,

TABLE IV
WDS AND EPS RESILIENCE QUANTITY SUMMARY

System	Independent Quantity	Interdependent Quantity
EPS	Bus Voltage Magnitude Line Thermal Limits Load (overall) supply satisfied	Thermoelectric cooling water demand satisfied Load (WDS pump) supply satisfied
WDS	Nodal Pressure	Pump Power Availability Zonal Demand Satisfaction

thermal limit, cooling water, demand supplied (overall), and demand supplied (WDS pumps) IOR values

$$R_{EPS,TOT}(t) = \left(w_v R_{R_V^n,t} + w_{tl} R_{R_{TL}^t,t} + w_{cw} R_{R_{CW}^g,t} + w_{ds} R_{R_{DS}^n,t} + w_{pl} R_{R_{PL}^n,t} \right) / (w_v + w_{tl} + w_{cw} + w_{ds} + w_{pl}). \quad (38)$$

VII. COMBINED CALCULATIONS

A. Combined IOR Computation

For the two interdependent infrastructure systems, the calculation of the IOR terms compose what are recognized as quantities pertaining solely to the operation of that particular system (EPS or WDS), $f(R_{IND})$, as well as quantities concerned explicitly with the interdependent operation of the two systems, $f(R_{INTR})$. Table IV summarizes the independent and interdependent quantities. The total resilience for the two systems at each time period can be calculated as follows:

$$\begin{aligned} R_{TOT}(t) &= f(\text{Re}_{SEPS,IND}) + f(\text{Re}_{SEPS,INTR}) \\ &+ f(\text{Re}_{SWDS,IND}) + f(\text{Re}_{SWDS,INTR}) \\ &= f(\text{Re}_{SIND}) + f(\text{Re}_{SINTR}) \\ &= \left(w_1 \left(R_{R_V^n,t} + R_{R_{TL}^t,t} + R_{R_{DS}^n,t} + R_{R_P^n,t} \right) + w_2 \left(R_{R_{CW}^g,t} + R_{R_{PL}^n,t} + R_{R_{Pump}^{pump,tim},t} + R_{R_{Dem}^{z,tim},t} \right) \right) / (w_1 + w_2) \end{aligned} \quad (39)$$

where the weights for the interdependent (w_2) and independent (w_1) terms can be determined through sensitivity analysis or by a decision maker.

B. Quantification Framework for IOR

Resilience of interdependent systems, such as the WDS and EPS, can be quantified using four metrics [21], including system robustness (40), resourcefulness of the operational control (43), recovery rapidity and recovery ability (41) and (42), and adaptability. A quantity also used to capture these behaviors is the time averaged performance loss (44). The plot in Fig. 4, modified from [21], shows a hypothetical resilience measure of performance value (IOR) and denotes the key points that will be used in the quantifying metrics below.

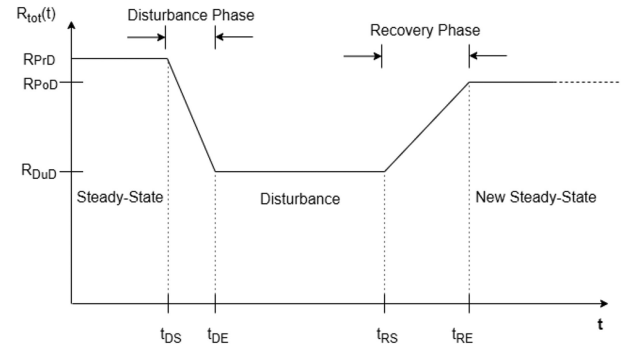


Fig. 4. IOR versus time (modified from [21]).

Robustness of a system is the measure of capability of its absorptive capability. Robustness in terms of IOR is given by the minimum value of IOR between the time at which the disturbance phase starts, t_{DS} , and when the recovery ends, t_{RE} , as follows:

$$M_{ROB} = \min(R_{TOT}(t), t_{DS} < t < t_{RE}). \quad (40)$$

Rapidity and performance loss are metrics used to quantify the disruption caused to the system performance during a certain disruptive phase. Rapidity is computed using measures in both the disturbance phase and the recovery phase and is defined using the average slope of the IOR. The performance loss is defined as the total reduction in IOR value during the disruption phase

$$M_{RAPI,DP} = \frac{R_{PrD} - R_{DuD}}{t_{DE} - t_{DS}} \quad (41)$$

$$M_{RAPI,RP} = \frac{R_{PoD} - R_{DuD}}{t_{RS} - t_{RE}} \quad (42)$$

$$M_{RA} = \left| \frac{R_{PoD} - M_{ROB}}{R_{PrD} - M_{ROB}} \right| \quad (43)$$

$$M_{TAPL} = \frac{\int (R_{PrD} - R_{TOT}(t)) dt}{t_{RE} - t_{DS}}. \quad (44)$$

Finally, the total resilience metric integrates the above-mentioned five metrics (40)–(44)

$$M_{TOT} = M_{ROB} \cdot \frac{M_{RAPI,RP}}{M_{RAPI,DP}} \cdot \frac{M_{RA}}{M_{TAPL}}. \quad (45)$$

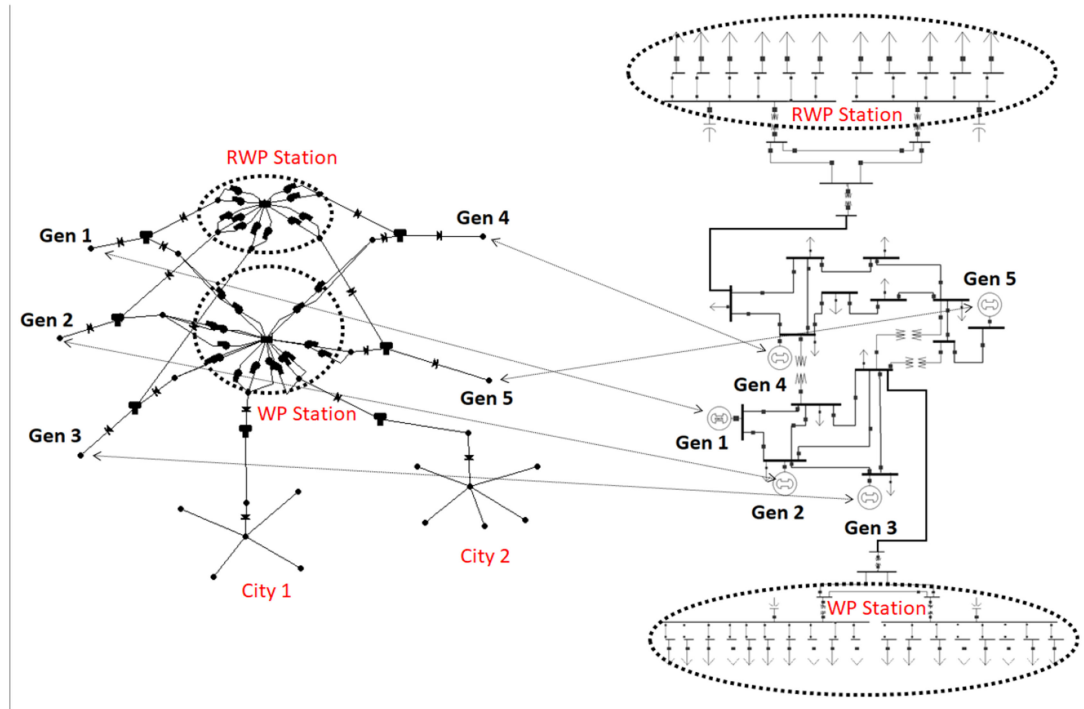


Fig. 5. Spatial relationship between EPS and WDS.

VIII. APPLICATION EXAMPLE

A. Example EPS

The example power system was a modified version of the IEEE 14 bus system [35], which is shown in Fig. 5. The modifications included the addition of representative distribution systems [36] to which the water system pumping station loads were mapped. Data for the distribution systems were generated from the IEEE 34-bus test distribution system [37]. The approach taken here is equivalent to modeling the voltages for voltage levels at and above the subtransmission system for all of the system, except where the critical interdependencies of the power system with the water system lie. Fig. 5 shows how the one-line-diagram of the 14-bus system and the two additional distribution systems were mapped spatially to the water system and vice versa. In order to create time varying system loads, historical load data [38] were normalized and used as LMF within the test system. The short-term security constrained UC for each hour was run every 24 h during the simulation using the next 24 LMF's, which had their indices offset according to the current time period in the simulation. Fig. 6 shows the LMF pattern that was used for the power system. Given the extreme drought scenario that is being simulated, an LP model that takes into consideration generation unit operational constraints, recent cooling water demands, and onsite storage tank level for determining the current maximum active power output of a given unit was utilized.

B. Example WDS

The example water system is a hypothetical WDS modeling demands from two cities and the power system described above.

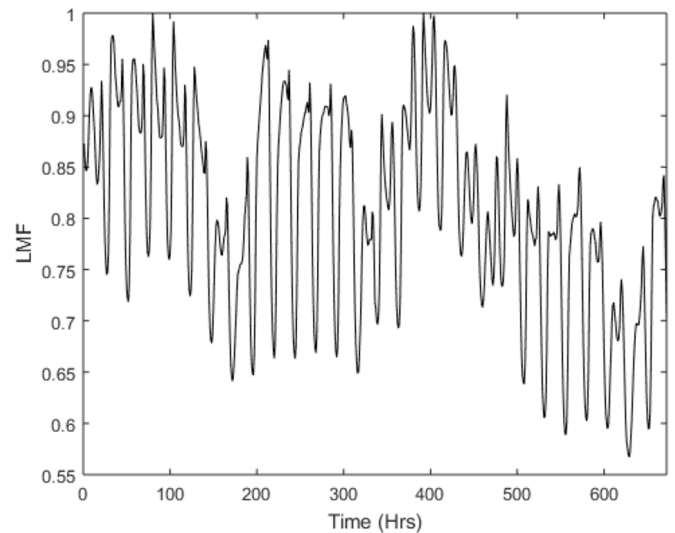


Fig. 6. LMF for one-month simulation.

Cooling water is supplied to the power plants in the WDS from both a freshwater source and a reclaimed water source (shown in Fig. 5). This cooling water is pumped to each power plant through a freshwater pumping station and a reclaimed water pumping station. The WDS control optimization-simulation model attempts to maximize the supply of reclaimed water for power plants. City 1 has four service areas with a total base demand of 30 000 gpm. City 2 has five service areas with a total base demand of 25 000 gpm. A total of 17 freshwater pumps (WP) and 11 reclaimed water pumps (RWP) serve the overall WDS. Cooling water demands for power plants, on an hourly

TABLE V
PUMPS AFFECTED BY EPS CONTINGENCY SCENARIO

	Pump Name	WDS Node	EPS Bus
1	RWP2.1	1	22
2	RWP2.2	2	23
3	RWP2.3	3	24
4	WP1.1	6	36
5	WP1.2	8	37

TABLE VI
NUMERICAL RESULTS FOR IOR QUANTIFICATION METRICS

Disturbance #	M_{ROB}	$M_{RAPI,DP}$	$M_{RAPI,RP}$	M_{RA}	M_{TAPL}	M_{TOT}
1	0.7261	0.0037	0.0021	0.2789	10.0893	0.0114
2	0.7296	0.0022	0.0004	0.3386	9.6786	0.0046
3	0.6917	0.0009	0.0043	2.3291	3.4585	2.2256

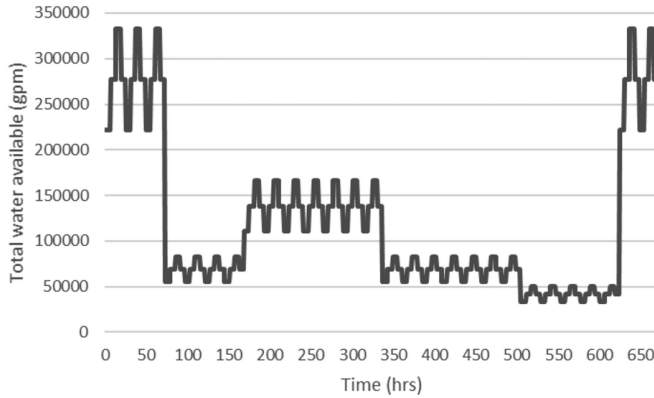


Fig. 7. Trends of water availability.

basis, are input to the WDS optimization–simulation model from the EPS optimization–simulation model. Each power plant in the system is equipped with onsite water storage equivalent to a two-week average water consumption.

C. Application Scenario

The scenario considered for the example application has a combination of water and power contingencies. The water contingency experienced by the water system is representative of drought conditions that would result in limited water availability for extended periods of time (> 2 weeks). Fig. 7 shows the trends in the system level water availability. The power contingency under consideration consists of several extended periods (black sky events) where two pumping stations (freshwater pumping station at power plant 1 and reclaimed water pumping station for power plant 2) experienced an outage. Table V contains the summary of the pumps that experienced the power contingency, whereas Fig. 8 shows the hours during which the power outages at pumping stations occurred.

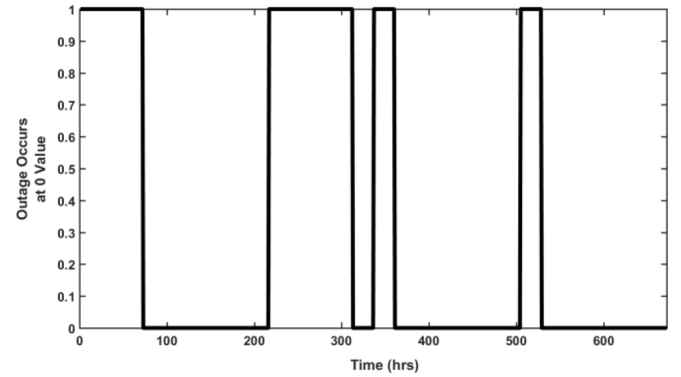


Fig. 8. Pump outage hours.

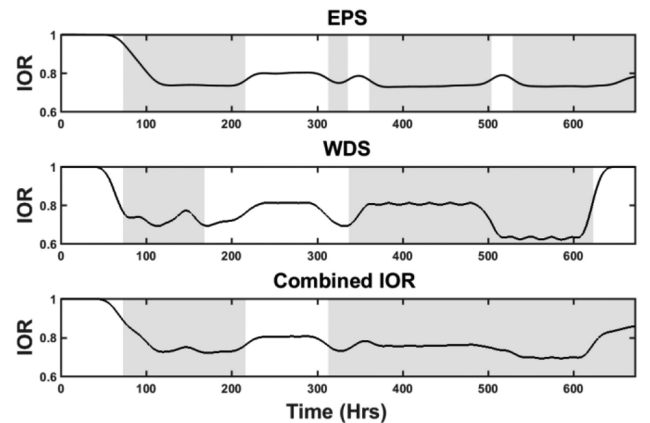


Fig. 9. IOR for EPS and WDS, combined with contingency overlay.

IX. RESULTS

Fig. 9 shows the trends of WDS IOR [using (33)], EPS IOR [using (38)], and combined IOR [using (39)] for both the independent and interdependent functions. The shaded areas in

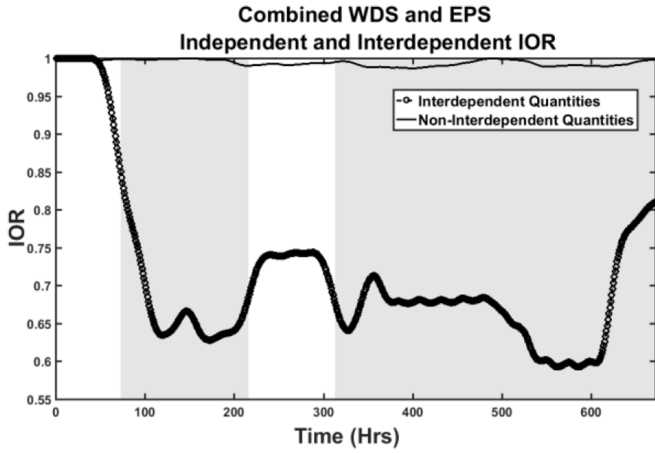


Fig. 10. Independent and interdependent IOR quantities versus time.

the WDS IOR graph show the contingency periods due to limited water availability (see Fig. 7), whereas the shaded areas in the EPS IOR graph show the contingency periods due to power outages of the pumps shown in Table V. The shaded areas in the combined IOR graph depict contingency periods due to limited water and/or power outage during the total simulation period of 672 h. The values of weights used for the independent (w_1) and interdependent functions (w_2) were 1 and 5, respectively. The trends of EPS and WDS IOR are observed to decline from simulation times slightly before the contingency start time of 72 h due to the data processing of the IOR (a moving average) used to postprocess the results. The lag observed in the decline of EPS IOR compared with WDS IOR is due to the EPS being more resilient to limited water availability on account of the onsite cooling water storage at the power plants. The EPS and WDS IOR follow similar trends during the later phases of the simulation. The WDS is observed to recover from the reduced IOR values at the end of the simulation period even though pump power outages still persist. This is because during the last 48 h of the simulation, there is no shortage of available water. Thus, it is noted that the effect of the pump power outages is mitigated in the WDS as the simulation progresses.

Fig. 10 shows the independent, $f(\text{Res}_{\text{IND}})$, and interdependent, $f(\text{Res}_{\text{INTR}})$, IOR values for the 672-h simulation of combined operations of WDS and EPS with normalizing weights on all terms equal to one. The shaded areas in the figure depict contingency periods due to limited water availability and/or power outages. Decrease in the interdependent IOR value was observed during the contingency periods, whereas the decrease in independent IOR value is seen to be much smaller, in comparison, throughout the simulation. It is also noted that the effect of the second round of severe contingencies beginning in hour 336 on the IOR values is not as severe as the effect of the first contingency that occurs in hour 72, demonstrating that the implemented control strategies effectively attenuate the impact of the second contingency and manoeuvre the interdependent systems to the new system operating conditions prior to hour 336. The benefit this article offers in considering separate computations for independent and interdependent IOR quantities with appropriate weights is evident from the fact that the decline in the IOR value captured by the interdependent IOR quantities

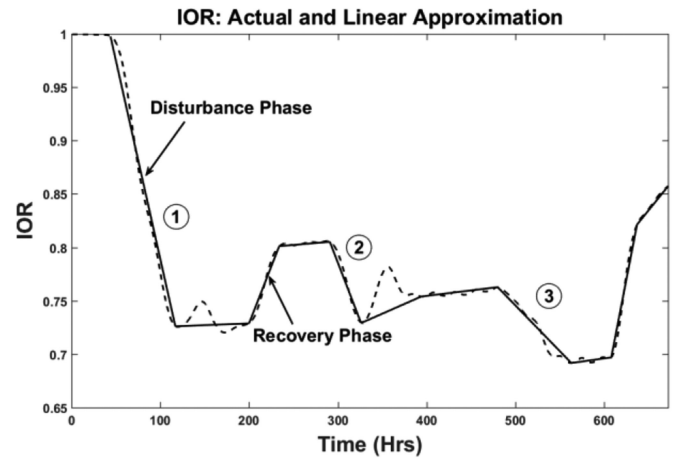


Fig. 11. IOR with linear approximation for metric application.

is not observed in the independent IOR quantities. A sensitivity analysis was performed for the weights of independent and interdependent IOR quantities, which allows for the consideration of the effect that each quantity has on the overall IOR value. It was observed that increasing the interdependent IOR weight (w_2) beyond ten has a diminishing effect on the overall resilience value (with the independent IOR weight w_1 remains constant) as the final IOR value is dominated by the interdependent terms, $f(\text{Res}_{\text{INTR}})$.

Fig. 11 shows the final IOR value, (39), using interdependent quantities weight $w_2 = 5$ and independent quantities weight $w_1 = 1$, along with a linear approximation of that curve. The linear approximation and accompanying visuals are shown to aid in understanding the application of the metrics shown in (40)–(45). Three disturbance periods are identified due to the contingency scenarios used in both networks. The encircled numbers in the plot draw attention to the beginning of each period, with the disturbance and recovery phase for the first disturbance period explicitly shown. Table VI contains the numerical values for the quantities discussed in the quantification framework section. The low recovery phase slope, captured by $M_{\text{RAPI,RP}}$, in disturbance period 2 as compared with disturbance period 1 results in the lower M_{TOT} for disturbance period 2. The relative decrease in the new steady-state value of IOR (M_{RA}) for disturbance period 2 as compared with 1 explains the increase in this value for disturbance 2. The lower time averaged performance loss, M_{TAPL} , for period 2 as compared with 1 is due to reasons just mentioned, and the larger value of M_{TAPL} for period 3 as compared with periods 1 and 2 results from the final IOR value being greater than the initial value. Finally, even though period 3 has a lower minimum IOR value, M_{ROB} , the faster recovery in IOR, $M_{\text{RAPI,RP}}$, combined with the much larger M_{RA} and much lower M_{TAPL} values result in the drastic difference in final, M_{TOT} .

X. CONCLUSION

The performance of the WDS and EPS under conditions of limited water availability and/or electric power outages is an important consideration for the design and operation of both systems, especially as the significance of considering their

interdependent operation becomes critical in such scenarios. This article presented a mathematical formulation for resilience calculations of the two systems, which accounts for operational and infrastructural considerations for each of the two systems as well as identified interdependencies in their operation. The new methodology for resilience computation, which has both of these considerations, were named IOR.

The IOR computational method presented in this article provided for the inclusion of IOR calculations within a previously developed extended time-period integrated EPS–WDS simulation optimization framework. The IOR computation methodology (shown in Fig. 1) was applied to a regional level WDS coupled with a corresponding EPS. Results of the application of this computation methodology for the example systems show the ability to capture system resilience when modeling short-term and long-term electric power outages and drought scenarios. This optimization–simulation framework used in conjunction with the methodology for IOR computation would allow utilities the ability to analyze real-time performance and facilitate decision making. In addition, the IOR methodology could be applied to forecasted, extreme, megadrought scenarios for more accurate system simulations and anticipatory policy making. The IOR computation methodology presented in this article does not consider component failures in either system in the computation procedure.

Because the resilience calculations for the test systems shown are not computationally expensive (total calculation time of 1.37s for EPS + 1.90s for WDS = 3.27s total), it seems that the IOR computation methodology could be used for both postsimulation as well as real-time EPS–WDS performance assessment. Therefore, future work will involve application of the developed IOR computation methodology to larger WDS and EPS test systems. The scalability of the proposed metrics is currently being investigated for larger test systems and will be incorporated in future research. Moving the application toward larger, more realistic test systems also warrants the inclusions of a developed middleware architecture, which emulates the supervisory control and data acquisition (SCADA) communication that would be necessary for data exchange between the two networks.

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